Chapter 2 Answers

2.4

Note that P_1 seems to be being used not just as an implication, but as an equivalence. See for example the proof of Lemma 4.3, where $\{a, b\} = \{a, d\}$ is said to imply $b \in \{a, d\}$. The equivalence is of course the definition that makes much more sense (if we only had the implication, we couldn't actually discern what $a = b$ even meant).

Induction with A4

Consider the infinite sequence $a_0, a_1, a_2, ..., a_n, ...$ such that $a_0 = \emptyset$ and for each n, $a_{n+1} = \{a_n\}$. We prove that if $n \neq m$, $a_n \neq a_m$. The base case is when $n = 0$, that is, $a_n = \emptyset$. Obviously, $\emptyset \neq a_m$ for every $m \neq 0$, as every a_m where $m > 0$ is a set that contains an element, that is, a non-empty set. Now suppose the result holds for some n, that is, if $n \neq m$, then $a_n \neq a_m$. Now consider a_{n+1}. We prove that if $n+1 \neq m$, then $a_{n+1} \neq a_m$. By similar reasoning as in the base case, $a_{n+1} \neq \emptyset$. Now consider any $m > 0$. By Note 3, if $a_{n+1} = a_m$, then $a_n = a_{m-1}$, as $a_{n+1} = \{a_n\}$ and $a_m = \{a_{m-1}\}$. Now, if $n + 1 \neq m$, then $n \neq m - 1$, and by the induction assumption, $a_n \neq a_{m-1}$, which is a contradiction. Thus $a_{n+1} \neq a_m$. If $n + 1 = m$, the statement vacuously holds.

Theorem 4.4

Note that this theorem can obviously be extended to equivalence, that is, $\langle a, b\rangle = \langle c, d\rangle$ iff $a = c$ and $b = d$.

n-tuples

One can easily prove by induction that this scheme of n-tuples works. The base case is just Theorem 4.4. Now suppose the result holds for some n, that is, for all sets $x_1, ..., x_n$ and $y_1, ..., y_n$, if $\langle x_1, ..., x_n\rangle = \langle y_1, ..., y_n\rangle$ then for every $i \leq n$, $x_i = y_i$. Now, if

$$\langle x_1, ..., x_n, x_{n+1}\rangle = \langle y_1, ..., y_n, y_{n+1}\rangle$$

then

$$\langle\langle x_1, ..., x_n\rangle, x_{n+1}\rangle = \langle\langle y_1, ..., y_n\rangle, y_{n+1}\rangle$$
$$\langle x_1, ..., x_n\rangle = \langle y_1, ..., y_n\rangle$$

and

$$x_{n+1} = y_{n+1}$$

by Theorem 4.4 and then, by the induction assumption, for every $i \leq n$, $x_i = y_i$. Thus for all $i \leq n+1$, $x_i = y_i$. Note that one could also take $\langle x_1, ..., x_n, x_{n+1} \rangle$ to be $\langle x_1, \langle x_2, ..., x_n, x_{n+1} \rangle \rangle$, as can be proven in a similar manner as above.

In fact, we can generalize the definitions of n-tuples in the following way. For $n = 2$, define it in the same way as before. Then for every $n > 2$, we could take $\langle x_1, ..., x_n, x_{n+1} \rangle$ to be any m-tuple (where $m < n$) of the form $\langle x_1, ..., x_a, \langle x_{a+1}, ..., x_b \rangle, x_{b+1}, ..., x_c \rangle$ where, if $c > 0$, then $c = n$, and if $c = 0$, then $b = n$. This scheme can be proven to work by the following induction. Because $n = 2$ is defined in the same way, the base case holds. Now suppose the result holds for all number less than n. Now consider any scheme for n of the form:

$$\langle x_1, ..., x_a, \langle x_{a+1}, ..., x_b \rangle, x_{b+1}, ..., x_c \rangle$$

Suppose that

$$\langle x_1, ..., x_a, \langle x_{a+1}, ..., x_b \rangle, x_{b+1}, ..., x_c \rangle = \langle y_1, ..., y_a, \langle y_{a+1}, ..., y_b \rangle, y_{b+1}, ..., y_c \rangle$$

Because both $\langle x_1, ..., x_a, \langle x_{a+1}, ..., x_b \rangle, x_{b+1}, ..., x_c \rangle$ and $\langle y_1, ..., y_a, \langle y_{a+1}, ..., y_b \rangle, y_{b+1}, ..., y_c \rangle$ are some m-tuples where $m < n$, the induction assumption holds, and the three following things are true: (1) for all $i \leq a$, $x_i = y_i$; (2) $\langle x_{a+1}, ..., x_b \rangle = \langle y_{a+1}, ..., y_b \rangle$; (3) for all $b+1 \leq i \leq c$, $x_i = y_i$. By (2) and the induction assumption, (1′) for all $a+1 \leq i \leq b$, $x_i = y_i$. Thus, combining (1), (1′), and (3) together, we have, for all $i \leq c$, $x_i = y_i$, that is, for all $i \leq n$, $x_i = y_i$, thus completing the proof.

Exercise 4.1

(a): Suppose $\langle x, y \rangle$ is defined to be $\{\{\emptyset, x\}, \{\{\emptyset\}, y\}\}$. Now suppose that $\langle a, b \rangle = \langle c, d \rangle$. Thus

$$\{\{\emptyset, a\}, \{\{\emptyset\}, b\}\} = \{\{\emptyset, c\}, \{\{\emptyset\}, d\}\}$$

meaning $\{\emptyset, a\} \in \langle c, d \rangle$. Thus either $\{\emptyset, a\} = \{\{\emptyset\}, d\}$ or $\{\emptyset, a\} = \{\emptyset, c\}$.

In the first case, we would have $\emptyset \in \{\{\emptyset\}, d\}$, meaning $\emptyset = d$. Then also $a = \{\emptyset\}$. Then we also have two sub-cases: either $\{\{\emptyset\}, b\} = \{\emptyset, \{c\}\}$ or $\{\{\emptyset\}, b\} = \{\{\emptyset\}, d\}$ (as $\{\{\emptyset\}, b\} \in \langle c, d \rangle$). In the first sub-case, we would have $b = \emptyset$ and $c = \{\emptyset\}$, and because $a = \{\emptyset\}$ and $d = \emptyset$, the result would be proven. In the second sub-case we would have $\{\{\emptyset\}, b\} = \{\{\emptyset\}, d\}$ (meaning $b = d$ by Lemma 4.3) and thus

$$\{\{\emptyset, a\}, \{\{\emptyset\}, b\}\} = \{\{\emptyset, c\}, \{\{\emptyset\}, d\}\} = \{\{\emptyset, c\}, \{\{\emptyset\}, b\}\}$$

that is,

$$\{\{\{\emptyset\}, b\}, \{\emptyset, a\}\} = \{\{\{\emptyset\}, b\}, \{\emptyset, c\}\}$$

and by Lemma 4.3 $\{\emptyset, a\} = \{\emptyset, c\}$, meaning $a = c$ (again by Lemma 4.3), thus proving the theorem.

In the second case, we have $\{\emptyset, a\} = \{\emptyset, c\}$ (meaning $a = c$ by Lemma 4.3) and thus $\{\{\emptyset\}, b\} = \{\{\emptyset\}, d\}$ by Lemma 4.3 and, again by Lemma 4.3, $b = d$, thus completing the proof.

(b): Now suppose that $\langle x, y \rangle$ is defined to be $\{\{\emptyset, \{x\}\}, \{\{y\}\}\}$. Now suppose that $\langle a, b \rangle = \langle c, d \rangle$. Thus $\{\emptyset, \{a\}\} \in \langle c, d \rangle$, meaning either $\{\emptyset, \{a\}\} = \{\{d\}\}$ or $\{\emptyset, \{a\}\} = \{\emptyset, \{c\}\}$. The first sub-case is ruled out, as then we would have $\emptyset = \{\{d\}\}$, which is impossible, as $\emptyset$ contains no elements, whereas $\{\{d\}\}$ contains an element. Thus $\{\emptyset, \{a\}\} = \{\emptyset, \{c\}\}$ meaning, by Lemma 4.3, $\{a\} = \{c\}$ and thus $a = c$. Then again by Lemma 4.3, $\{\{b\}\} = \{\{d\}\}$ and thus $b = d$, hence showing $a = c$ and $b = d$.

Now consider the second case, where $\{\emptyset, \{a\}\} = \{\emptyset, \{c\}\}$. Thus $\{a\} = \{c\}$ (by Lemma 4.3) and $a = c$, meaning, again by Lemma 4.3, $\{\{b\}\} = \{\{d\}\}$ and thus $b = d$, therefore completing the proof.

2.5

Notes

Note that if x is a set, then so is $\cap x$, as $\cap x \subseteq \cup x$, and $\cup x$ is a set by A_5.

Exercise 5.1

For suppose for any set x there is a set y that contains all elements of all elements of x. Now, $\cup x$ is defined to be the class of all elements of all elements of A. Thus $\cup x$ is a subclass of y (in fact, they are identical classes), and hence, by A_2, is a set.

Exercise 5.2

(a): Suppose $A \subseteq B$. Then, for all $x \in A$, $x \in B$. Now suppose $x \in \cup A$. Then $x \in y$ for some $y \in A$, that is, $y \in B$. Now, $\cup B$ is the class of all elements of all elements of B. But we have shown that x is an element of y, which is an element of B. Thus $x \in \cup B$. Hence $\cup A \subseteq B$.

(b): Suppose $A \subseteq B$. Then, for all $x \in A$, $x \in B$. Now suppose that $x \in \cap B$. Then $x \in y$ for every $y \in B$, and thus $x \in z$ for every $z \in A$. But $\cap A$ is the class of all elements of all elements of A, and we have shown that $x \in z$ for every element z of A. Hence $x \in \cap A$. Thus $\cap B \subseteq \cap A$.

Exercise 5.3

(a): This is true. For suppose that A is transitive. Then all elements of all elements of A are elements of A. $\cup A$ is the class of all elements of all elements of A. Now suppose that $x \in \cup A$. Thus, because it is an element of some element of A, we know that x is an element of A by the transitivity of A. Hence $x \in A$, and therefore $\cup A \subseteq A$.

(b): This is true. For suppose that $\cup A \subseteq A$. Then all elements of all elements of A are elements of A, that is, A is transitive.

(c): This is true. For suppose that A is transitive. Then all elements of all elements of A are elements of A. Now, $\cup A$ is the class of all elements of all elements of A. But suppose $\cup A$ was not transitive, that is, there was some element x of some element y of $\cup A$ such that $x \notin \cup A$. Now, we know that, because y is an element of some element of A, and because A is transitive, $y \in A$. But this means that x is an element of an element of A, that is, $x \in \cup A$. Hence $\cup A$ is transitive.

(d): This is true. For suppose that every element of A is transitive. Then, for every element x of A, all elements of all elements of x are elements of x. Now suppose that $\cup A$ was not transitive. Hence there would be some element x of an element y of $\cup A$ such that $x \notin \cup A$. Because $y \in \cup A$, y is an element of some element z of A. Because z is transitive, all elements of all elements of z are elements of z. But $x \in y$; hence $x \in z$. Thus x is an element of an element of A, meaning $x \in \cup A$. Thus $\cup A$ is transitive.

Exercise 5.4

(a): On the one hand, suppose that $z \in x \cup y$. Then $z \in x$ or $z \in y$. To be an element of $\cup\{x, y\}$, z must be an element of either x or y. But we have shown this. Hence $z \in \cup\{x, y\}$. On the other hand, suppose that $z \in \cup\{x, y\}$. Then z is either an element of x or y. But this means $z \in x \cup y$. Hence, because $z \in x \cup y$ iff $z \in \cup\{x, y\}$, $x \cup y = \cup\{x, y\}$.

(b): On the ond hand, suppose that $z \in x \cap y$. Then $z \in x$ and $z \in y$. To be an element of $\cap\{x, y\}$, z must be an element of both x and y. But we have shown this. Hence $z \in \cap\{x, y\}$. On the other hand, suppose that $z \in \cap\{x, y\}$. Then $z \in x$ and $z \in y$, that is, $z \in x \cap y$. Hence $x \cap y = \cap\{x, y\}$.

Exercise 5.5

(a): On the one hand, suppose that $y \in \cup\{x\}$. Then y is in some element of $\{x\}$. Because x is the only such element, $y \in x$. On the other hand, suppose that $y \in x$. Thus y is in some element of $\{x\}$, namely x, meaning $y \in \cup\{x\}$. Hence $\cup\{x\} = x$, meaning the statement is necessarily true.

(b): On the one hand, suppose that $y \in \cap\{x\}$. Then y is an element of all elements of $\{x\}$. But x is the only such element, meaning $y \in x$. On the other hand, suppose that $y \in x$. Then y is in all elements of $\{x\}$, as x is the only such element, meaning $y \in \cap\{x\}$. Hence $\cap\{x\} = x$, meaning the statement is necessarily true.

Exercise 5.6

(a): On the one hand, suppose that $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. Hence either $x \in B$ or $x \in C$. If $x \in B$, then $x \in A \cap B$. If $x \in C$, then $x \in A \cap C$. Thus either $x \in A \cap B$ or $x \in A \cap C$, meaning $x \in (A \cap B) \cup (A \cap C)$. On the other hand, suppose that $x \in (A \cap B) \cup (A \cap C)$. Then either $x \in A$ and $x \in B$, or $x \in A$ and $x \in C$. Thus $x \in A$ and either $x \in B$ or $x \in C$, that is, $x \in A \cap (B \cup C)$.

(b): On the one hand, suppose that $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$. If $x \in A$, then x is in both $A \cup B$ and $A \cup C$, and hence also in $(A \cup B) \cap (A \cup C)$. If $x \in B \cap C$, then $x \in B$ and $x \in C$. Then $x \in A \cup B$ and $x \in A \cup C$, meaning $x \in (A \cup B) \cap (A \cup C)$. On the other hand, suppose that $x \in (A \cup B) \cap (A \cup C)$. Then $x \in A \cup B$ and $x \in A \cup C$. If $x \in A$, then $x \in A \cup (B \cap C)$. If $x \notin A$, then $x \in B$ and $x \in C$, that is, $x \in B \cap C$, meaning $x \in A \cup (B \cap C)$.

(c): On the one hand, suppose that $x \in A - (A - B)$. Then $x \in A$ and $x \notin A - B$. Thus either $x \notin A$ or $x \in B$. But it cannot be that $x \notin A$, for it was assumed that $x \in A$. Thus $x \in B$. Hence $x \in A$ and $x \in B$, meaning $x \in A \cap B$. On the other hand, suppose that $x \in A \cap B$. Then $x \in A$ and $x \in B$. Hence $x \notin A - B$. Thus $x \in A$ and $x \notin A - B$, meaning $x \in A - (A - B)$.

(d): The question gives the incorrect statement to prove. It should instead say $B - (A - B) = B$, not $\emptyset$. Now, $x \in B - (A - B)$ iff $x \in B$ and $x \notin A - B$. Thus either $x \notin A$ or $x \in B$. Thus this statement is satisfied for all $x \in B$. But suppose there were some $x \notin B$ such that $x \in B - (A - B)$. But that is impossible, as x must be in B. Hence $x \in B$ iff $x \in B - (A - B)$, therefore meaning $B - (A - B) = B$.

(e): On the one hand, suppose that $A \subseteq B$. Now suppose that $x \in A \cap B$. Then $x \in A$. Now alternatively suppose that $x \in A$. But because $A \subseteq B$, $x \in B$. Thus $x \in A \cap B$. Thus $A \cap B = A$. On the other hand, suppose that $A \cap B = A$. Now suppose that $x \in A$. Thus $x \in A \cap B$, meaning $x \in B$. Hence $A \subseteq B$.

(f): On the one hand, suppose that $A \subseteq B$. Now suppose that $x \in A \cup B$. Then $x \in A$ or $x \in B$. If $x \in A$, then, because $A \subseteq B$, $x \in B$. In the second case, $x \in B$. Thus $x \in B$. Now alternatively suppose that $x \in B$. Thus $x \in A \cup B$. Thus $x \in A \cup B$ iff $x \in B$, meaning $A \cup B = B$. On the other hand, suppose that $A \cup B = B$. Now suppose that $x \in A$. Then $x \in A \cup B$, and thus $x \in B$.

(g): Suppose that $B \subseteq A$. By part (c) of this problem, $A - (A - B) = A \cap B$. By part (e) of this problem, and the assumption that $B \subseteq A$, $B \cap A = B$. But $B \cap A$ is just $A \cap B$. Thus $A \cap B = B$, meaning $A - (A - B) = B$.

2.6

Remark 1

Here is why Remark 1 is true. For suppose we instead had the axiom A_6': for any set x there is a set y that contains all subsets of x as elements. Now, the class $\mathcal{P}(x)$ contains all of those and only those subsets of x. Thus $\mathcal{P}(x)$ is a subset of (and equal to) the set that exists by A_6'. Hence, by A_2, $\mathcal{P}(x)$ is also a set.

Exercise 6.1

(a): Suppose that $y \in x$. Now, $\mathcal{P}(\cup x)$ is the set of all subsets of $\cup x$, and $\cup x$ is the set of all elements of all elements of x. Thus $\mathcal{P}(\cup x)$ is the set of all sets whose only elements are elements of elements of x. Now, either y is empty, or it is non-empty. If it empty, then $y \in \mathcal{P}(\cup x)$ vacuously holds. On the other hand, if it is non-empty, then y is a set of elements of an element of x, namely y itself. Thus y is a subset of $\cup x$, and hence $y \in \mathcal{P}(\cup x)$. Thus we have shown that $x \subseteq \mathcal{P}(\cup x)$.

(b): On the one hand, suppose that $y \in \cup(\mathcal{P}(x))$. What this means is that y is an element of an element of $\mathcal{P}(x)$. But all elements of $\mathcal{P}(x)$ are subsets of x, that is, they only contain elements of x. Hence y is an element of x. On the other hand, suppose that $y \in x$. Thus y is an element of an element of $\mathcal{P}(x)$, namely any subset of x which contains y. Hence $y \in \cup(\mathcal{P}(x))$. Thus $\cup(\mathcal{P}(x)) = x$.

(c): Suppose that $x \subseteq y$. Now suppose that $z \in \mathcal{P}(x)$. Thus z is a subset of x, that is, z is such that all of its elements are elements of x. But that means all of the elements of z are also elements of y. Thus all elements of z are elements of y, meaning z is a subset of y, and thus $z \in \mathcal{P}(y)$. Hence $\mathcal{P}(x) \subseteq \mathcal{P}(y)$.

2.7

Exercise 7.1

On the hand hand, suppose that $x \in (A \cup B) \times (C \cup D)$. Thus $x = \langle a, b \rangle$ for some $a \in A \cup B$ and $b \in C \cup D$. Thus there are four cases: $a \in A$ and $b \in C$; $a \in A$ and $b \in D$; $a \in B$ and $b \in C$; $a \in B$ and $b \in C$. This is the same as saying either $x \in A \times C$ or $x \in A \times D$ or $x \in B \times C$ or $x \in B \times D$. That is,

$$x \in (A \times C) \cup (A \times D) \cup (B \times C) \cup (B \times D)$$

On the other hand, suppose that

$$x \in (A \times C) \cup (A \times D) \cup (B \times C) \cup (B \times D)$$

Thus $x = \langle a, b \rangle$ for some a either in A or B and for some b either in C or D, that is,

$$x \in (A \cup B) \times (C \cup D)$$

Hence the statement is true.

2.8

Exercise 8.1

Suppose that $x \in \text{Dom}(R)$. Thus there is some y such that $\langle x, y \rangle \in R$. Thus $\{\{x\}, \{x, y\}\} \in R$. Also $\{x, y\} \in \{\{x\}, \{x, y\}\}$, meaning $\{x, y\} \in \cup R$. But $x \in \{x, y\}$. Thus $x \in \cup(\cup R)$. Hence $\text{Dom}(R) \subseteq \cup(\cup R)$.

2.10

Notes

Note that the reason Theorem 10.4 and Theorem 10.5 only hold for sets is that $\mathcal{P}(x)$ was only defined for sets (not for classes in general).

Note that Theorem 10.3 is just Exercise 5.3 (d).

In Theorem 10.5, here is why $x \subseteq \mathcal{P}(x)$ implies that $\mathcal{P}(x) \subseteq \mathcal{P}(\mathcal{P}(x))$. Because $x \subseteq \mathcal{P}(x)$, every element of x is a subset of x. But now suppose that $y \in \mathcal{P}(x)$. Then y is a subset of x. Now, $\mathcal{P}(\mathcal{P}(x))$ is the set of all subsets of all subsets of x. But, for every element of x, that element is a subset of x. Thus, because y is a set containing only elements of x, all elements of y are elements of $\mathcal{P}(x)$. But that means y is a subset of $\mathcal{P}(x)$, that is, $y \in \mathcal{P}(\mathcal{P}(x))$.

One can generalize the results obtained in the first three paragraphs in this section. Let the sequence $a_1, a_2, ..., a_n, ...$ be defined in the following way: $a_1 = \emptyset$, $a_2 = \{a_1\}$, $a_3 = \{a_1, a_2\}$, and in general, $a_{n+1} = \{a_1, ..., a_n\}$. We prove that every a_n is transitive. The base case was explained in Note 2 of Section 3. Now suppose the result holds for all number less than $n + 1$, that is, all sets a_i for $i \leq n$ are transitive. Now, $a_{n+1} = \{a_1, ..., a_{n-1}, a_n\}$. We can prove that all elements of all elements of a_{n+1} are elements of a_{n+1} by limited induction. The case for a_1 is vacuously true. Now suppose the result holds for all numbers less than some $m < n$. That is, all elements of a_i are in a_{n+1} for every $i < m$. Now, $a_m = \{a_1, ..., a_{m-1}\}$. But $a_1, ..., a_{m-1}$ are all in a_{n+1}, that is, all elements of a_m are in a_{n+1}. Thus all elements of all elements of a_{n+1} are in a_{n+1}.

Another generalization can be made (this time of paragraph two). Suppose we considered some set $x = \{a_m, ..., a_n\}$ where $m < n$, but such that the set does not have n elements. This is just the $\{a_1, ..., a_n\}$ but with at least one a_i removed (for $i < n$). We prove that this set cannot be transitive. Let a_k be (at least) one of the elements missing from the set $\{a_m, ..., a_n\}$. Now, $a_n \in x$. But $a_n = \{a_1, ..., a_k, ..., a_{n-1}\}$, that is, $a_k \in a_n$. But $a_k \notin x$. Thus there is an element of x that has an element that is not in x, meaning x is not transitive.

Here is an explanation for why every a_n defined in the preceeding two paragraphs really is a set. a_1 and a_2 have already been shown to be sets. Now, $a_3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$. Using A_4, we know that

$$x = \{\{\emptyset, \{\emptyset\}\}, \{\{\emptyset, \{\emptyset\}\}\}\}$$

is a set. By A_5, we know that $\cup x$ is a set. But $\cup x = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$, that is, $\cup x = a_3$, meaning a_3 is a set by A_2. Now suppose the result holds for some for all numbers less than n (where $n \geq 2$). Now, $a_n = \{a_1, ..., a_{n-2}, a_{n-1}\}$. By the induction assumption, $\{a_1, ..., a_{n-2}\}$ is a set, as $\{a_1, ..., a_{n-2}\} = a_{n-1}$, which was assumed to be a set. Then, by A_4,

$$x = \{\{a_1, ..., a_{n-2}\}, \{a_{n-1}\}\}$$

is a set. Thus, by A_5, $\cup x$ is a set. But

$$\cup x = \{a_1, ..., a_{n-2}, a_{n-1}\} = a_n$$

Thus, by A_2, a_n is a set, completing the proof.

It should be noted that the results in the above paragraphs also hold if one instead defined $a_1, ..., a_n, ...$ as we did previously: $a_1 = \emptyset$ and $a_{n+1} = \{a_n\}$, and then considered the same sequence of sets: $c_1 = \emptyset$ and $c_{n+1} = \{a_1, ..., a_n\}$. It should also be noted that these results prove that there are an infinite number of transitive sets. Here is the proof that $\{a_1, ..., a_n\}$ when defined in this way is still a set (for every n). We can show the more general fact that, for any sets $x_1, ..., x_n$, $\{x_1, ..., x_n\}$ is a set. The base case (when $n = 1$) holds by A_4. Now suppose that the result holds for some n, that is, $\{x_1, ..., x_n\}$ is a set for all sets $x_1, ..., x_n$. Now consider $\{x_1, ..., x_n, x_{n+1}\}$ (for any sets $x_1, ..., x_n, x_{n+1}$). By A_4, and the assumption that $\{x_1, ..., x_n\}$ is a set, $\{\{x_1, ..., x_n\}, \{x_{n+1}\}\}$ is a set. Then, by A_5, $\cup\{\{x_1, ..., x_n\}, \{x_{n+1}\}\}$ is a set. But $\cup\{\{x_1, ..., x_n\}, \{x_{n+1}\}\} = \{x_1, ..., x_n, x_{n+1}\}$, thus showing $\{x_1, ..., x_n, x_{n+1}\}$ is a set. As a consequence of this result, $\{a_1, ..., a_n\}$ is a set for all n.

For the sake of clarity, let us refer to the sequence of a_i's in the last paragraph as b_i. That is, $b_1 = \emptyset$ and $b_{n+1} = \{\emptyset\}$ and $c_1 = \emptyset$ and $c_{n+1} = \{b_1, ..., b_n\}$. Let us refer to the previous defintion of a_i's in the same way. That is, $a_1 = \emptyset$ and $a_{n+1} = \{a_1, ..., a_n\}$. Interestingly enough, b_i is never swelled (if $i > 2$), but c_i is always swelled, whereas a_i is swelled for $i \leq 3$, but is not swelled for any $i > 3$. First, we show the former statement: that b_n is swelled for $i \leq 2$, but is not swelled for $i > 2$, and c_i is always swelled. The base cases (for $i = 1$ and $i = 2$) hold, as it has already been shown that $\emptyset$ and $\{\emptyset\}$ are supercomplete (and hence swelled). Now consider any b_n for $n > 1$. Then $b_n = \{b_{n-1}\}$. This obviously means b_n is not swelled, as a subset of b_n is $\emptyset$, but that is not a member of $\{b_{n-1}\}$, unless $b_n = \{\emptyset\}$, in which case $n = 2$. Now consider c_n, that is, $\{b_1, ..., b_n\}$. Now, for each $i > 1$, the subsets of b_i are $\emptyset$ and b_{i-1}, and there are no subsets of the empty set. But c_n includes b_{i-1} and the empty set. Thus c_n contains every subset of each of its elements. Now we consider a_n. Both a_1 and a_2 have been shown to be supercomplete (hence swelled). Consider $a_3 : \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$. The subsets of all of its elements are: $\emptyset, \{\emptyset\}$ and $\{\emptyset, \{\emptyset\}\}$. But a_3 has all of these as elements. Thus it is swelled. But now consider any a_n for $n > 3$. Then a_3 is an element of a_n. One subset of a_3 is $x : \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$. But $x \notin a_n$, as $x \neq a_m$ (as can be easily shown) for any $m < n$ (and $a_n = \{a_1, ..., a_{n-1}\}$). Hence there is a subset of an element of a_n which is not an element of a_n, meaning a_n is not swelled.

It should be noted that the above case for c_n assumes the following (which I am not sure is implied by the definitions given in the book): Suppose we have some subset y of x, and some set $z = y$. Then, if $z \in x$, then $y \in x$. Or, more generally, for any two sets y and z, if $z = y$ and $z \in x$, then $y \in x$. Here is why this relates to the above. For suppose we considered some $c_n : \{b_1, ..., b_n\}$. Then we would have, for example, $\{b_1, b_1\} \subseteq c_n$. But we do not explicitly have $\{b_1, b_1\} \in c_n$, only $\{b_1\} \in c_n$. But $\{b_1\} = \{b_1, b_1\} \in c$. Thus, if the first statement of this paragraph held, then we would also have $\{b_1, b_1\} \subseteq c_n$. It seems that this is how the book is using the notion of set membership. Recall Note 2 of section 4. It is stated here that $a = b$ means a and b are the same element. Thus, because $\{b_1, b_1\} = \{b_1\}$, they are the same element. Hence $\{b_1, b_1\} \in c_n$.

Here is why $x : \{\emptyset, \{\emptyset, \{\emptyset\}\}\} \notin a_n$ for every n. We can prove a more general statement: Any set of the form $x : \{x_k, ..., x_n\}$ is not equal to any set of the form $y : \{x_j, ..., x_m\}$ such that y contains at least one x_i such that $i \neq l$ for all $x_l \in x$. A statement that we will see implies this is: $a_n \neq a_m$ for all n and m such that $n \neq m$. The base case, where $n = 1$, is obvious, as $a_1 = \emptyset$, whereas all a_m for $m > 0$ have at least one element. Now suppose the result holds for all numbers $\leq n$. Now consider a_{n+1}, and any a_m such that $m \neq n$. Either $m > n$ or $m < n$. In the first case, $a_{n+1} = \{a_1, ..., a_n\}$ and $a_m = \{a_1, ..., a_n, ..., a_m\}$. Then it must be that $a_m = a_i$ for some $i \leq n$ (if $a_{n+1} = a_m$). But, becuase $m \neq i$ for all $i \leq n$, it follows by the induction assumption that $a_m \neq a_i$ for all $i \leq n$. Thus $a_m \notin a_{n+1}$, meaning $a_{n+1} \neq a_m$. On the other hand, suppose that $m < n$. Then it follows that $a_m = \{a_1, ..., a_{m-1}\}$ where $m - 1 < n$. Then, if $a_{n+1} = a_m$, $a_n = a_i$ for some $i < n$. But this is not possible by the induction assumption. Thus $a_n \notin a_m$, meaning $a_{n+1} \neq a_m$.

Now having proven this result, it is obvious that the other result holds: Any set of the form $x : \{x_k, ..., x_n\}$ is not equal to any set of the form $y : \{x_j, ..., x_m\}$ such that y contains at least one x_i such that $i \neq l$ for all $x_l \in x$. For suppose this was the case. Then x_i is such that $i \neq l$ for all $x_l \in x$. But, by the last result, it then follows that $x_i \neq x_l$ for all $x_l \in x$. Thus $x_i \notin x$, meaning $x \neq y$.

Exercise 10.1

(a): The only element of $\{\{\emptyset\}\}$ is $\{\emptyset\}$. It's only element is $\emptyset$. But $\emptyset \notin \{\{\emptyset\}\}$. Thus $\{\{\emptyset\}\}$ is not transitive. In fact, $\{\{\emptyset\}\}$ is not transitive as a consequence of the result of the paragraph preceeding the last.

(b): Suppose that we removed some element x from V. Thus the resulting class is $V - \{x\}$. Now, because $x \in V$, $\{x\} \in V$. But we didn't remove $\{x\}$ from V. Thus $\{x\} \in V - \{x\}$. But $x \notin V - \{x\}$. Thus there is some element of $V - \{x\}$ who has an element that is not in V, meaning $V - \{x\}$ is not transitive.